

Who trades against the maker? Adverse selection with counterparty identity on an on-chain options order book[★]

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ABSTRACT

A market maker on an options book loses to counterparties who know something and earns from those who do not, but on an anonymous venue the two can only be told apart by proxy. Derive settles on chain, so its public tape names both wallets. On 621 508 fills in BTC, ETH and HYPE options between January 2024 and September 2026, the maker's mean markout thirty minutes after the fill is +12.79 USDC per contract: a half spread of +15.45 plus adverse selection of -2.66. Fees, rebates and a modelled hedge cost leave a net edge of +9.34. The loss is extremely concentrated: ten taker wallets, almost the entirety of the two professional counterparty classes, carry 90.1 per cent of it. Trade size, the standard proxy for toxic flow, explains none of it once instrument-by-day fixed effects are included, and sweeps only the move of the underlying, which hedging removes. Four hypotheses were registered before any markout was computed; three are rejected.

1. Introduction

A market maker on an options book loses money to counterparties who know something and earns it from those who do not. Telling the two apart is the central problem of the literature that follows Glosten and Milgrom (1985) and Copeland and Galai (1983). Because the counterparty is anonymous, toxicity has to be inferred from what is observable about the trade itself, above all its size and its aggressiveness (Easley and O'Hara, 1987). That inference is indirect, and it is the weakest link in any estimate of what an options maker actually earns.

Derive is an options venue whose book is matched off chain and settled on chain. Every fill lands in a public tape that names both wallets, the subaccount, the request-for-quote identifier where there is one, the fee, the rebate and the realised profit of each side. The counterparty is not a statistical construct but an address with a history. This paper uses that to ask directly what the literature has had to approach by proxy: against whom does an options maker lose? In equity options the question has been answered through the price impact of net demand (Bollen and Whaley, 2004;

Gârleanu, Pedersen and Poteshman, 2009) and through the returns that follow order flow (Muravyev, 2016), always with the counterparty unobserved (Easley, O'Hara and Srinivas, 1998).

The answer is that the maker's average result is a statement about the counterparty rather than about the option. Taking the passive side of all 621 508 fills in the sample, the mean markout after thirty minutes is +12.79 USDC per contract, while the median fill returns 0.96. Splitting that average by counterparty class moves it from +24.46 against ordinary takers to -25.40 against the seventeen wallets that dominate maker volume. Nothing about the option changes across that range; only the address on the other side does.

The loss is carried by very few of them. Ten taker wallets account for 90.1 per cent of the maker's aggregate loss, and a single address for 38.0 per cent. The same ten wallets supply 98.8 per cent of the fills of the dominant-maker class and 92.0 per cent of those of the market-maker programme, so the concentration result and the counterparty result are one finding seen twice. What does not follow is the usual explanation. Once instrument-by-day fixed effects absorb the composition of the flow, trade size above the ninetieth percentile carries no coefficient distinguishable from zero, and sweeps across several strikes carry one only in USDC, through the next move of the underlying, which a delta hedge removes. On this venue the standard observables do not find the informed trader; the identity does.

Three contributions follow. The first is a measurement. Section 3 decomposes the maker's result into a half spread, adverse selection, the fee, the rebate and the cost of delta hedging, marked against the volatility surface that the venue itself writes on chain rather than against a mark of my own construction. The second is the counterparty fact above, together with its negative half: the observables that the literature uses in place of identity do not reproduce it. The third, exploratory, is a map. Section 6 reports the net edge per unit of notional across moneyness and tenor for three

[★]First version 19 September 2026, revised 25 September 2026, both on pilot data to 17 September 2026; this version is computed on the registered sample to 30 September 2026, 08:00 UTC, and every number in it is recomputed. The four verdicts are those of the pilot, which held 97 per cent of these fills, so they are no independent replication. This version also reports the test of H1 in all three units, draws the placebo windows of H3 as registered and keeps the funding rate and the settlement markouts to the registered windows. The revision of 25 September corrected the unit of the maker fee and rebate, which the tape books as sums over the whole fill: the first version subtracted them from a markout per contract, divided the net edge per contract by the notional of the whole fill in its map, and mixed the same units in the markout as a share of the premium. On the pilot data the correction moved the net edge from +8.93 to +9.58 USDC per contract and the share of positive cells under H4 from 49.5 to 76.3 per cent, and redrew the map per notional; Addendum 3 to the pre-registration records it. The first version had also inverted the wording of H2, H3 and H4, not their tests.

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underlyings, which is the form in which a quoting decision is actually taken.

A fourth contribution is procedural. Four hypotheses, their rejection rules and the inference were committed to the repository before a single markout was computed (Burlig, 2018). Three of the four are rejected. They are reported as such, because a pre-registered rejection is evidence about the market and not a failure of the study: it is the reason the reader can believe the one result that survives.

2. The venue and the tape

Crypto markets are fragmented across venues with very different designs (Makarov and Schoar, 2020). Derive matches orders off chain and settles them on an Optimism-stack chain of its own. Positions are held in subaccounts under a portfolio margin system, and the mark that margin is computed against is not a private number. A feed contract receives, for every expiry, the parameters of a stochastic volatility inspired surface (Gatheral and Jacquier, 2014), and anyone can read the mark implied volatility of any strike at any block by evaluating that surface. During the sample the median curve was 32 seconds old at the moment of a fill and the ninetieth percentile was 58 seconds.

The public trade tape is unusual: it is not anonymous. Each fill appears twice, once from each side, and carries the wallet and subaccount of both parties, the transaction hash, the fee and rebate booked to each side, and the realised profit. Where a trade came from a request for quote, the tape carries the identifier that ties its legs together; 20.7 per cent of fills arrive this way.

The sample runs from 11 January 2024 to 30 September 2026 and covers the option fills on BTC, ETH and HYPE that pass the registered exclusions: 621 508 fills across 43 495 instruments, 12 019 taker wallets and 7 264 maker wallets. ETH carries 414 707 fills, BTC 159 467 and HYPE 47 334. Liquidations do not appear: they are settled outside the trade tape, so the liquidation class defined in the pre-registration is empty throughout.

Almost everything known about adverse selection in decentralised markets concerns automated market makers, where the liquidity provider does not quote but posts a curve, and where the loss to arbitrageurs is a mechanical consequence of that curve rather than a decision (Milionis, Moallemi, Roughgarden and Zhang, 2022; Cartea, Drissi and Monga, 2024; Lehar and Parlour, 2025). A central limit order book with a maker who chooses a price is a different object, closer to the venues studied by Hasbrouck and Sofianos (1993) and Menkveld (2013) than to a constant-product pool, and the evidence on decentralised market quality has so far been drawn from the pool side (Barbon and Rinaldo, 2026; Aramonte, Huang and Schrimpf, 2021). Prices on this venue are quoted by people, and what follows measures what those quotes earned.

What the venue does not publish is the history of the book itself. Quotes that were never hit, the depth standing

at any moment, and cancellations are absent, so every statement in this paper is about trades that happened and none is about the cost of quoting. Section 7 returns to this.

3. What a markout measures

For every fill I take the maker's side, $s = -1$ when the taker bought and $s = +1$ when the taker sold, and compare the traded price P with the venue's mark M at a series of horizons. The markout at horizon τ is $\text{MO}_\tau = s(M(t + \tau) - P)$. At $\tau = 0$ this is the half spread, the distance between the fill and the mark at the moment of the fill, and the difference $\text{MO}_\tau - \text{MO}_0$ is adverse selection: what the mark did afterwards. Figure 1 shows the construction on a single fill. The two pieces sum to the markout by construction, so the half spread must not be added a second time when the maker's result is totalled.

Three marks are available and they answer different questions. The primary one is the on-chain surface, evaluated at the last curve pushed at or before the horizon, because it is the number the venue itself uses for margin and settlement. The second is the mark attached to the next fill in the same instrument, which is a genuine transaction but arrives at a median distance of 15 011 seconds and is therefore no control at a thirty-minute horizon. The third is the settlement payoff, which is exact but only available once. The first two are compared in Figure 9; the settlement payoff enters the secondary test of H2.

Subtracting the maker fee, adding the maker rebate and subtracting a modelled cost of holding the resulting delta to the horizon turns the markout into a net edge. The tape books the fee and the rebate for the whole fill, so both are divided by its number of contracts before they enter. For a multi-leg request for quote the fee is almost always booked on one leg; spreading it over the package leaves the net edge at 9.34 and changes no cell verdict. Over the sample the fee costs 1.07 USDC per contract, the rebate returns 0.77 and the hedge costs 3.15, which takes the markout of +12.79 to a net edge of +9.34, with a 95 per cent interval from 1.22 to 17.95. Figure 2 shows the whole identity with an interval on every component. The hedge is the only component that is not observed, and the figure marks it as such; its assumed cost is varied in Section 5. The registered net edge charges the hedge to the markout in USDC, which still carries the move of the underlying; on the delta-neutral markout the same net edge is +11.75 (exploratory).

Four units are used and none of them is redundant. USDC per contract is the quantity the pre-registration fixed. Vol points remove the move of the underlying and pool across underlyings whose contracts differ in size by three orders of magnitude. The share of the premium is what a maker books. Basis points of notional is what a quoting decision compares against a perpetual spread. Figure 3 shows why reading the result in only one of them is misleading.

Counterparties are assigned to one class each by a cascade in which the first match wins: liquidation, vault, request for quote, dominant maker, market-maker programme, large

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BTC-20251018-109000-P maker sold 0.03 contracts

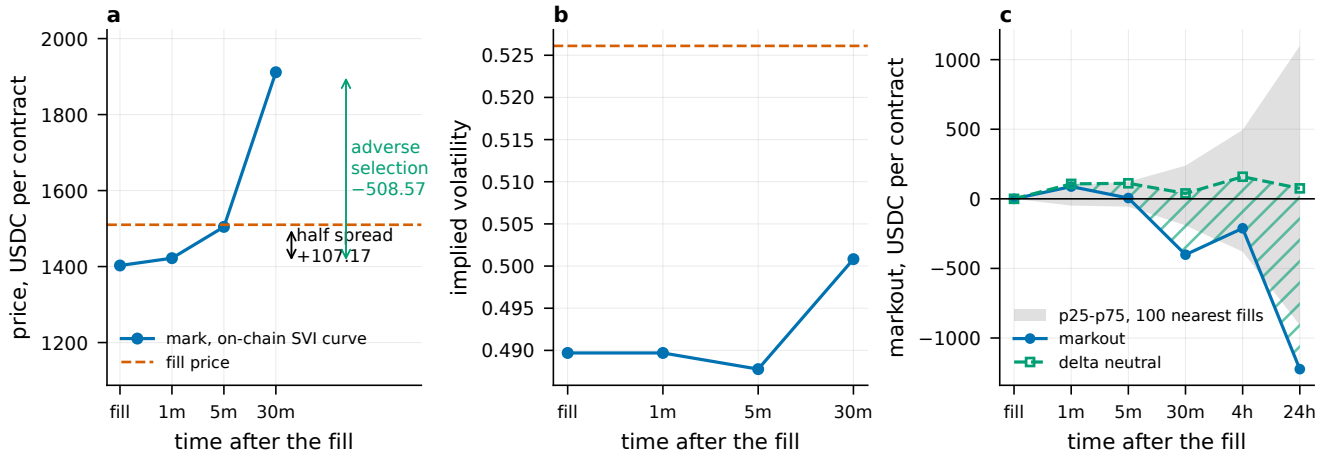


Figure 1: What a markout measures. Panels a and b stop at the registered 30 minute horizon, panel c carries every horizon. The shaded band is the interquartile range of the 100 nearest fills in the same cell, so the example can be read against its neighbours rather than on its own.

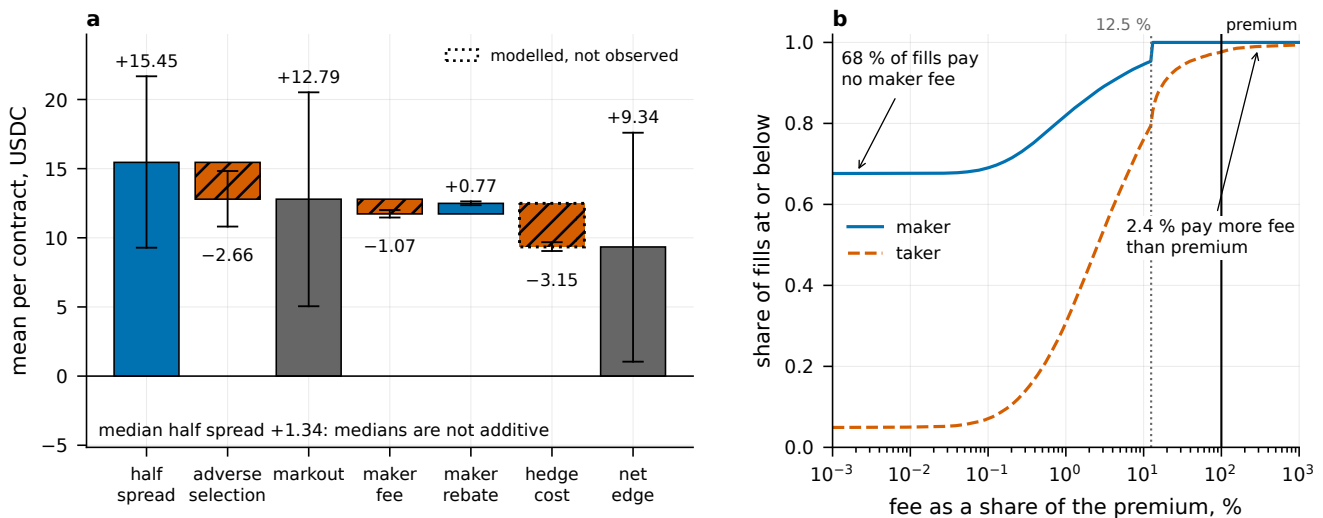


Figure 2: From the half spread to the net edge. Error bars in panel a are 95 per cent cluster bootstrap intervals over taker wallets with 999 draws. Fee and rebate in panel a are per contract, the amount booked for the fill divided by its size. The hedge bar is dotted because it is the only component that is modelled rather than observed. Panel b is the distribution of the fee of a fill as a share of its premium, for both sides of the trade.

wallet, and otherwise ordinary taker. A dominant maker is a wallet that supplied a large share of maker volume in the same month; the programme membership comes from the venue's own published epochs. Figure 5 shows the cascade with the fills it assigns.

4. Pre-registration and inference

Four hypotheses were committed, with their rejection rules, on 17 September 2026 before any markout existed. Two dated addenda resolved questions of definition while the data were being built; a third, written after the first version, corrects the unit of fee and rebate and changes no hypothesis, threshold or sample. H1 holds that toxic flow is concentrated in few wallets *and* identifiable from trade size and sweeps;

it is rejected unless the upper bound of its interval reaches one half and both coefficients are negative and significant; its coded test reads the delta-neutral markout. H2 holds that vault flow is uninformed; it is rejected if the mean markout against vaults is significantly negative. H3 holds that HYPE flow on Derive was more toxic before HYPE options were listed on a centralised venue on 23 June 2026 than after; it is rejected unless the difference-in-differences estimate is positive, significant, and extreme against a placebo distribution. H4 holds that the net edge is positive in fewer than half of the moneyness-by-tenor cells and negative in short-dated at-the-money BTC and ETH; it is rejected if half of the cells, or either of those, carry a positive interval.

All inference clusters on the taker wallet, the level at which the sampling here is plausibly independent (Abadie,

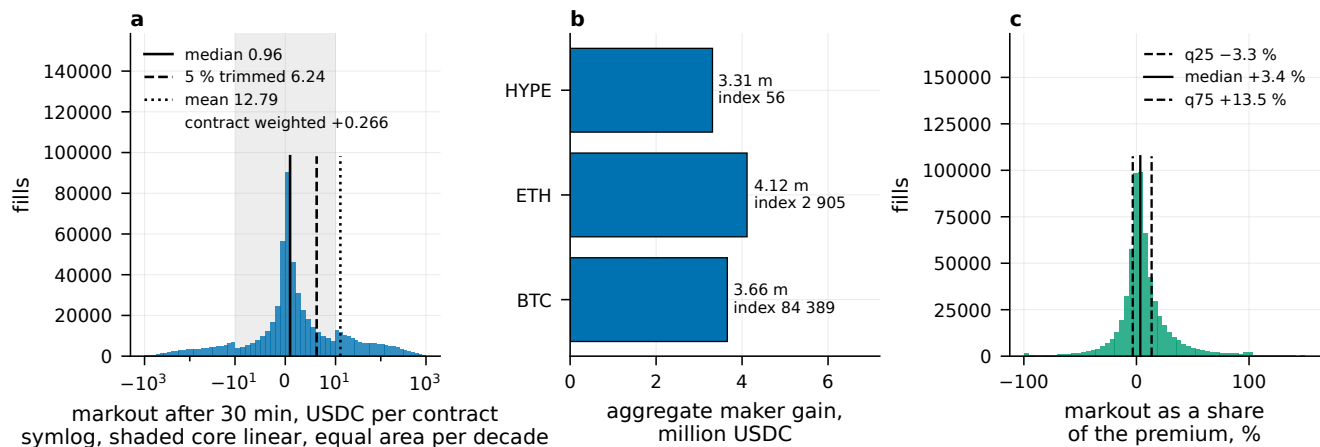


Figure 3: How to read the headline number. The shaded core of panel a is linear and everything outside it is logarithmic with equal area per decade. Panel b sets the aggregate dollar gain against the average index price, which is what makes a per-contract average a statement about contract size. Panel c divides the markout per contract by the price per contract.

Athey, Imbens and Wooldridge, 2023). Coefficients are tested with a wild cluster bootstrap using restricted residuals and Rademacher weights (Cameron, Gelbach and Miller, 2008; MacKinnon and Webb, 2017), with $B = 999$ draws and a fixed seed (Roodman, Nielsen, MacKinnon and Webb, 2019); means and cell averages carry percentile intervals from a cluster pairs bootstrap with the same B . Fixed effects, applied as a within transformation, are instrument by day with delta-bucket dummies for H1 and day for H3. Error bars and bands that appear only in a figure use 999 draws or fewer.

5. Results

5.1. The average is not a trading outcome

The mean markout is thirteen times the median, and the gap is not a curiosity but the first thing a reader has to absorb. Weighting each fill by the contracts it moved rather than counting it once takes the average from +12.79 to +0.266 USDC per contract; trimming five per cent from each tail takes it to +6.24. Figure 3 shows why. The three underlyings contribute similar aggregate dollar amounts to the maker, 3.66, 4.12 and 3.31 million USDC, while their average index prices differ by a factor of 1 520. A per-contract average is therefore in large part a statement about contract size. Expressed as a share of the premium, both taken per contract, the distribution a maker books runs from -3.3 per cent at the lower quartile through +3.4 per cent at the median to +13.5 per cent at the upper quartile. Every number that follows is reported in the unit that makes it comparable, and the pre-registered unit is kept alongside.

5.2. Adverse selection is small, and it does not grow

Restricting the sample to the 526 230 fills that have every horizon, so that a falling line cannot be a shrinking sample, the maker's median result in vol points is almost flat from one minute to one day, from 1.22 to 1.09, while in USDC per

contract it falls from 1.33 to 0.79; Figure 4 shows it in three units. Adverse selection, the markout less the half spread of +16.15, deepens in USDC from -0.48 at one minute to -2.48 at five minutes and -3.72 at one day. Delta-neutral it is -0.15 at thirty minutes, and in vol points it stays within 0.05 of zero up to four hours. What the counterparty learns is the next move of the underlying, most of it within five minutes, not the volatility. Splitting the flow makes the point sharper: the professional classes sit below zero at every horizon and the remaining flow above it, and the two curves are already separated at one minute. This is the first sign that the maker's problem is not information about volatility arriving after the trade.

5.3. The counterparty, not the option

Figure 5 reports the six classes with the number of wallets behind each. The ordering is stark, from +24.46 USDC per contract against ordinary takers and +14.85 against request-for-quote flow down to -20.47 against the market-maker programme and -25.40 against dominant makers. Against dominant makers the maker's half spread is already -16.59 before anything happens: the fill is struck on the wrong side of the mark. Adverse selection adds a further -8.80. Against the market-maker programme the split is -4.94 and -15.53. Against ordinary takers the half spread is +25.30 and adverse selection only -0.84. In vol points, which remove the move of the underlying, the median adverse selection is within 0.06 of zero for every class, and the classes differ in the half spread alone: -0.70 vol points against dominant makers, -0.55 against the market-maker programme and +2.04 against ordinary takers. The adverse selection in USDC against the programme is the next move of the underlying, which a delta hedge removes. The professional counterparties get a better entry price; they do not trade on information about volatility. After fees, rebate and hedge the net edge per class runs from +21.79 USDC per contract against ordinary takers to -31.97 against dominant makers. Against vault flow the markout is +4.21, with an

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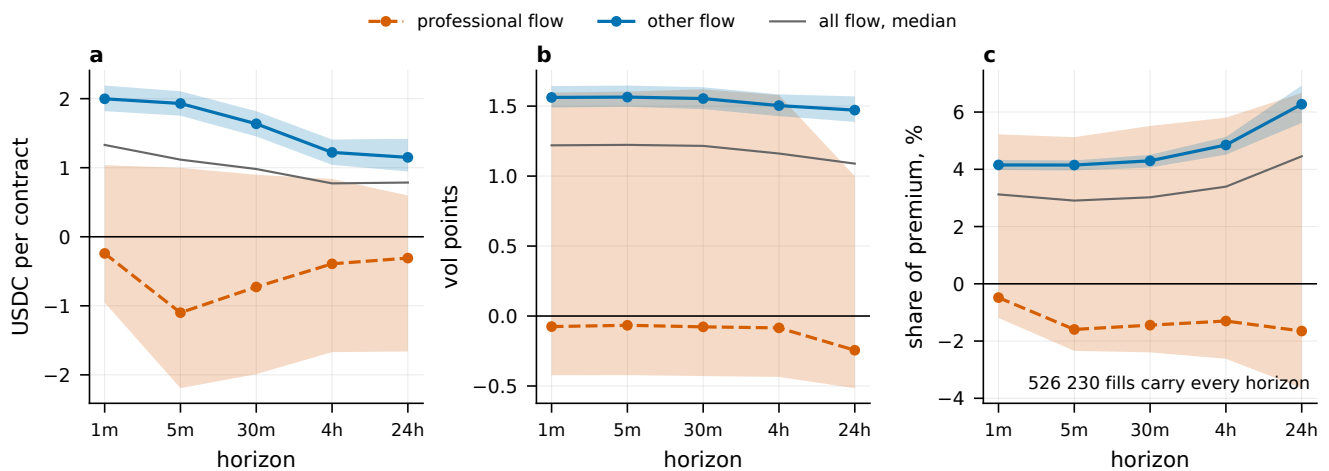


Figure 4: How long adverse selection lasts. The subsample is restricted to fills that have every horizon, so a falling line cannot be a shrinking sample. Bands are 95 per cent cluster bootstrap intervals of the median, not dispersion. The share of the premium in panel c is taken per contract.

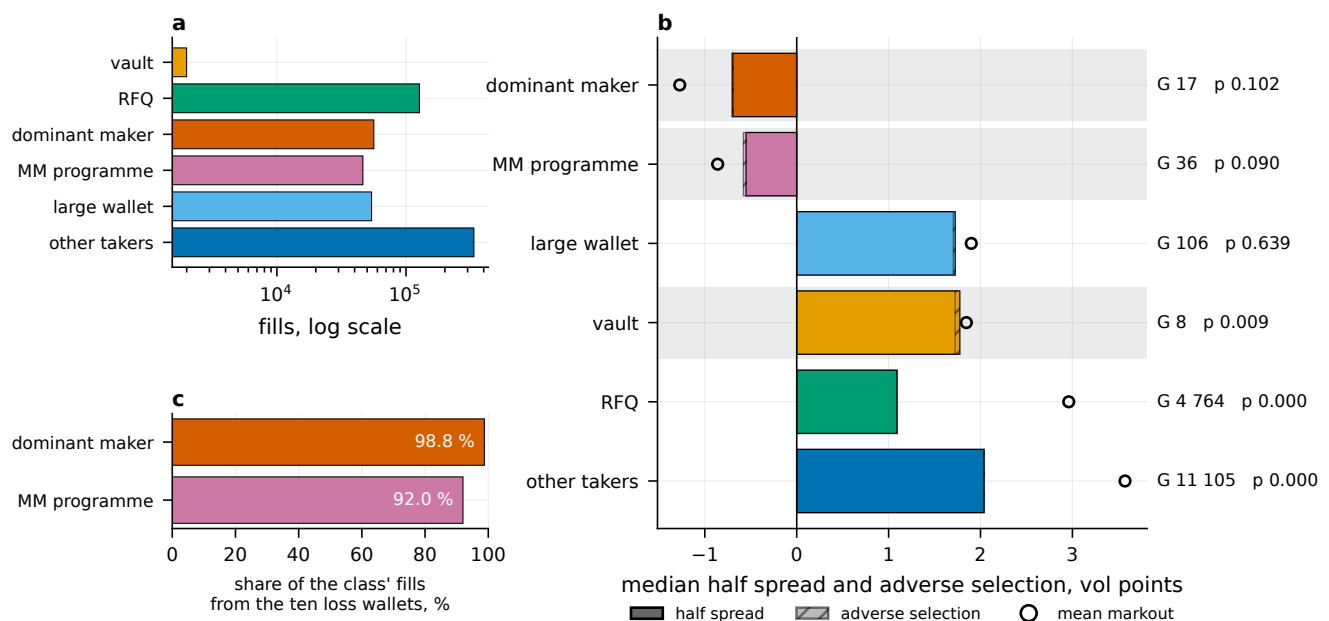


Figure 5: Who takes back which part of the spread. G is the number of taker wallets behind a class and p is the wild cluster bootstrap p-value against zero; shaded rows carry fewer than 40 wallets. Medians are shown per component and are not additive, so the mean markout is marked separately.

interval from 1.64 to 8.53 over eight wallets, so H2 is not rejected; the net edge is +2.08, given without an interval. The registered secondary test, on the settlement markout less its cell-month mean, does not reject either, at -6.37 with an interval from -17.78 to 5.11 ($p = 0.304$); its point estimate is negative.

Two cautions belong here. The two classes with negative markouts rest on seventeen and thirty-six wallets, and their wild cluster bootstrap p -values against zero are 0.102 and 0.090, above the five per cent level of the registered tests. The percentile intervals exclude zero and the p -values do not, so the class result is reported as a description and not as a test.

5.4. Concentration, and the failure of H1

Ten wallets carry 90.1 per cent of the maker's aggregate loss, summed per contract as registered, with an interval from 68.8 to 94.0 per cent that is wide because a single address carries 38.0 per cent of it on its own. Only 1 449 of the 12 019 taker wallets are loss-making for the maker at all. Figure 6 shows the curve with its interval and, beside it, the reason H1 fails. Raw differences look convincing: fills above the ninetieth size percentile and sweeps across strikes both show a markedly worse delta-neutral result for the maker. Under instrument-by-day fixed effects, on the delta-neutral markout that the registered test reads, the size coefficient is -0.74 with $t = -0.77$ and the sweep coefficient +0.65

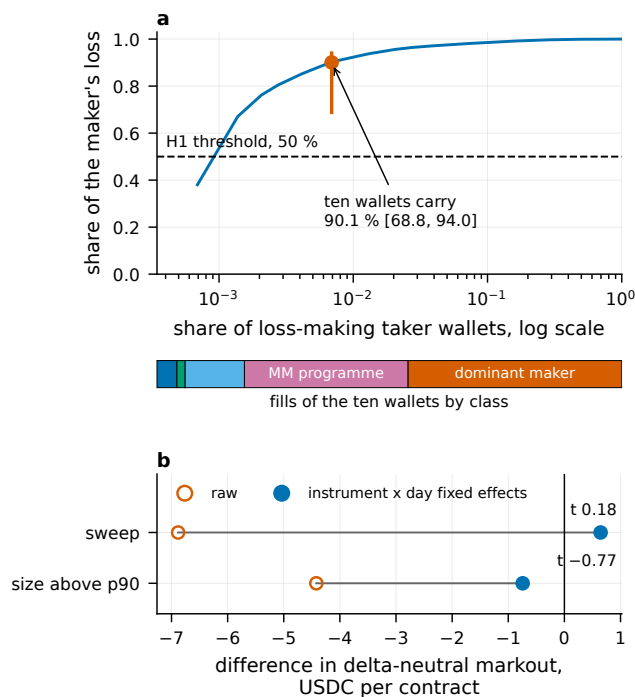


Figure 6: Concentration, and why size and sweeps explain nothing. The horizontal axis of panel a is logarithmic because the first ten wallets carry most of the loss. Panel b sets the raw difference against the coefficient under instrument by day fixed effects, which is where the hypothesis fails.

with $t = 0.18$; in vol points they are -0.15 ($t = -1.20$) and $+0.13$ ($t = 0.16$). In USDC per contract size is again insignificant, at -1.58 with $t = -0.93$, but a sweep costs the maker 13.54 ($t = -4.42$): it predicts the next move of the underlying, which a delta hedge removes. Concentration is real; the standard proxies for it are not. H1 is rejected because size fails in every unit and the hypothesis asserted both.

5.5. Where an edge survives, and the failure of H4

Of the 99 cells of underlying, moneyness and tenor with at least 200 fills, 71 carry a net edge whose 90 per cent interval lies above zero, none lies below it, and 28 are inconclusive. Twenty-three of those 28 sit in the rows with an absolute delta of 60 per cent or more, which carry 13 per cent of the fills. Figure 7 shows them. H4 asserted the opposite on both of its branches, and both fail. The share of positive cells is 71.7 per cent against a registered threshold of one half, and 76.8 and 56.6 per cent at an assumed perpetual half spread of zero and three basis points. Short-dated at-the-money options are positive in BTC at $+24.24$ USDC per contract, with an interval from 6.60 to 47.19, and in ETH at $+1.04$, from 0.19 to 2.11. At three basis points neither interval excludes zero, so there the rejection rests on the majority alone. One qualification remains. The rule reads percentile intervals, and the wild cluster bootstrap p -values of the two at-the-money cells are 0.117 and 0.169; counting only cells with a positive mean and p below 0.10 (exploratory) gives 55 cells, 55.6 per cent, but at three basis points only 47,

47.5 per cent, below the threshold. Read per unit of notional (exploratory), the edge is 2.5 to 9.6 times higher beyond ninety days than within two days in every underlying and every delta row below 75 per cent. That is the form Section 6 uses, and it is where the illiquidity premium documented for equity options would lead one to look (Christoffersen, Goyenko, Jacobs and Karoui, 2018).

5.6. One sample or two, and the failure of H3

The sample is not homogeneous and Figure 8 makes that checkable rather than assumed. HYPE enters only in November 2025 and the market-maker programme class cannot be assigned before November 2024, so every pooled average in this paper spans a market that changed composition underneath it. The difference-in-differences estimate for the listing of HYPE options on a centralised venue, over 90 days on either side, is -1.10 vol points with $t = -1.27$; 2400 fills of the window have no vol unit and are left out. Of 100 placebo windows of the same length, all before the listing, 52 per cent give a larger estimate in absolute value; the placebo estimates spread with a standard deviation of 0.50 vol points. The sign is wrong for the hypothesis and the magnitude is indistinguishable from noise drawn before the event; H3 is rejected for want of discrimination, not because the effect was shown to be absent.

6. What this means for a maker

A quoting decision is not taken in USDC per contract. It is taken per unit of risk, against the alternative of quoting the same notional somewhere else. Measured that way, net of fees, rebate and hedge, Figure 7 reads differently (exploratory). At the money the median rises from 2.8 basis points at two days or less to 5.6 at thirty to ninety days in BTC, from 3.8 to 10.0 in ETH and from 5.7 to 24.4 in HYPE. BTC, which pays most per contract, pays least per notional: its median over the cells is 3.2 basis points, against 5.7 in ETH and 15.6 in HYPE. The first consequence is that the margin is thinnest where the flow is. Options of up to seven days and an absolute delta below 60 per cent carry close to half of the BTC and ETH fills and pay 1.0 to 2.8 and 1.6 to 3.8 basis points respectively. That leaves little room for costs the model does not see: at a perpetual half spread of three basis points the share of positive cells falls to 56.6 per cent.

The second consequence concerns what a maker should spend effort on. The result of Section 5 is that size and aggressiveness carry no information about the outcome once the instrument and the day are held fixed, while the identity of the counterparty carries almost all of it. A maker who prices flow by its observable shape is modelling the wrong variable. A maker who maintains a list of a dozen addresses and widens against them is using the variable that works, and on a venue that settles on chain that list is cheap to build and needs no inference at all. The concentration makes the policy tractable: ten wallets, not thousands.

The third is a warning about the map itself. It describes fills that happened at prices that cleared. It does not describe

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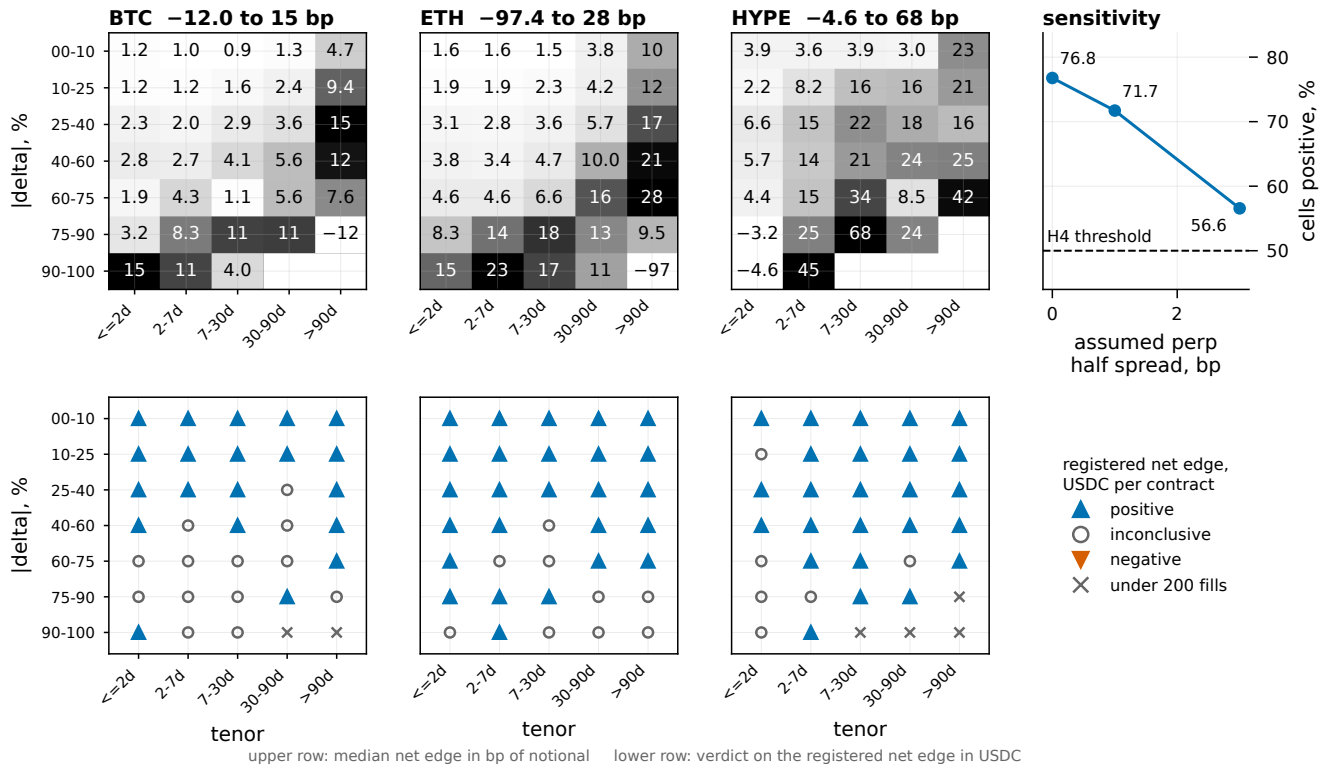


Figure 7: Where the edge survives. The upper row is the median net edge in basis points of notional, the edge of a fill over its notional and the unit a quoting decision uses, shaded within each panel only, between its 5th and 95th percentile. The lower row is the registered quantity in USDC and its verdict per cell, with an empty cross where a cell has fewer than 200 fills.

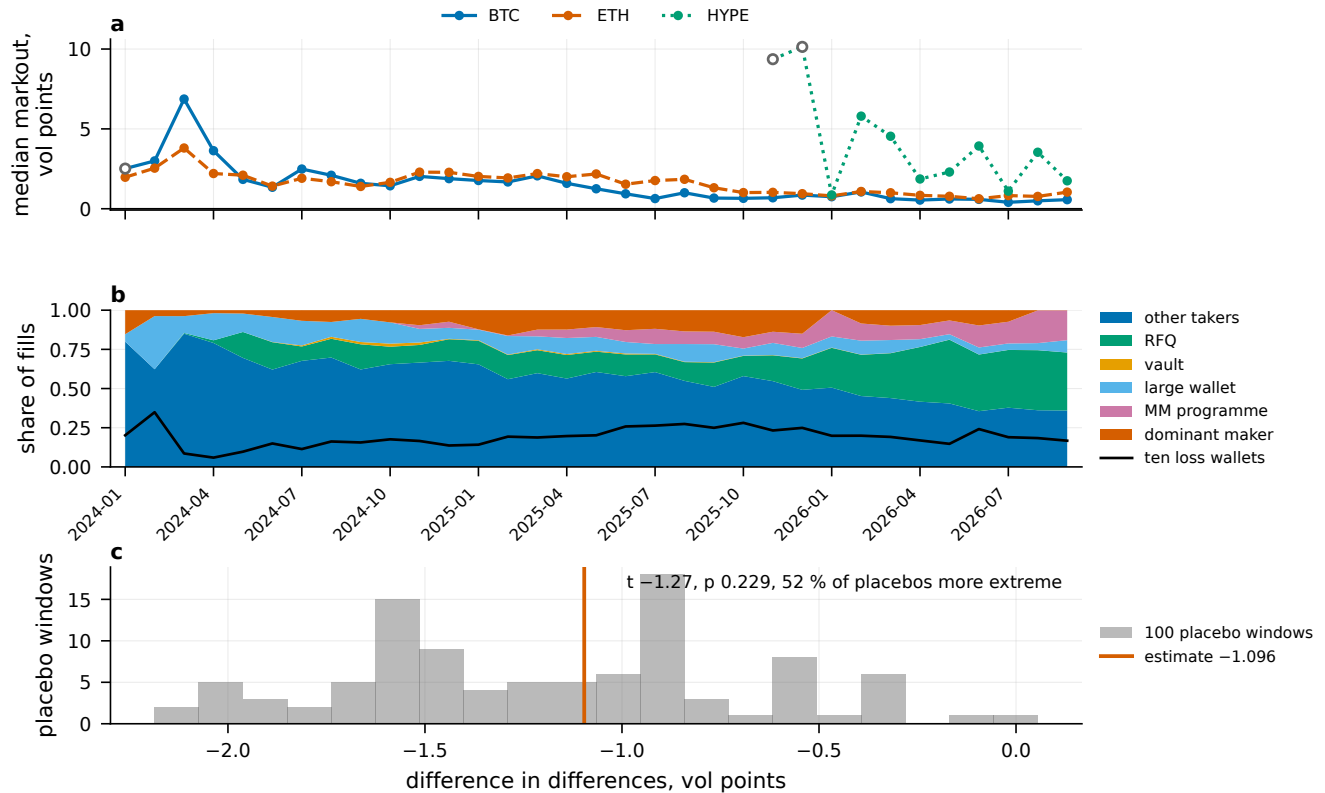


Figure 8: The sample over 33 months, and the event study. Panels a and b share a calendar axis; hollow markers mark months with fewer than 1000 fills. Panel c is the distribution of the 100 placebo estimates with the estimated effect marked.

Who trades against the maker?

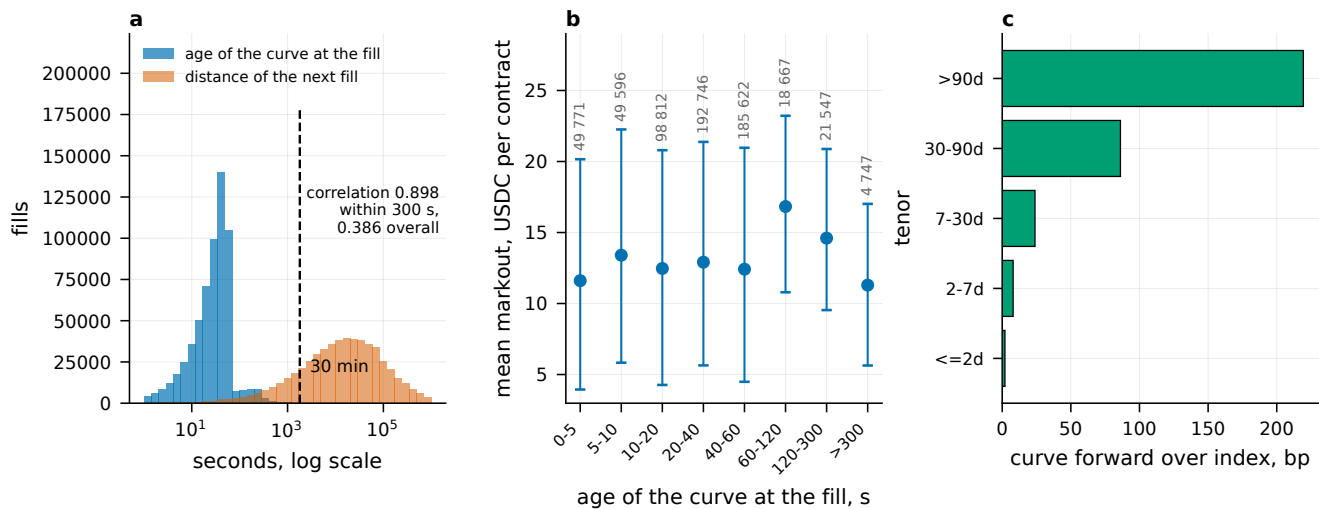


Figure 9: Is the on-chain mark good enough? Panel a puts the age of the curve and the distance of the next fill on one axis, which is why the next fill is no control at 30 minutes. A flat profile in panel b is the reassurance: the markout does not move with the age of the curve, and the fills per class stand above each point. Panel c is the mean excess of the curve's forward over the index by tenor.

what a new quote would have earned, because a quote that is never hit leaves no trace in the data. A cell with a large positive edge and few fills may be a cell where the maker was rarely willing to quote, not one where quoting was profitable. Sizing a strategy from this map without the book history would mistake selection for opportunity.

7. What this cannot show

Seven limits bound the reading of the results, and each is stated once. First, the mark is the venue's own surface, so the maker is measured against the number the venue uses rather than against an independent valuation; Figure 9 shows no trend in the mean markout across eight classes of curve age, from 11.30 to 16.83 USDC per contract within wide intervals (exploratory), so the lag of the curve does not drive the result. Second, the alternative mark taken from the next fill in the same instrument arrives at a median distance of 15011 seconds and is therefore not a control at a thirty-minute horizon; where the comparison is fair, with a next fill inside 300 seconds, the two paths correlate at 0.898 (exploratory). Third, one trader may hold several addresses, so the concentration figures in this paper likely understate it. Fourth, the vault class rests on eight wallets, so few that the bootstrap itself is fragile, which is why H2 is reported as a hint that vault flow is not toxic and not as a demonstration. Fifth, liquidations settle outside the trade tape, so the paper says nothing about the flow that a maker probably fears most. Sixth, the volatility unit is undefined for 11 059 fills whose traded price sits outside the arbitrage bounds, and those fills are excluded from every vol-point statement. Seventh, and most importantly, the venue publishes no history of the book, so nothing here measures the cost of quoting, only the outcome of trading.

8. Conclusion

On a venue where the counterparty is an address rather than an inference, the maker's result is mostly a statement about who was on the other side. Adverse selection after the fill is close to zero in vol points for every class; in USDC it is the underlying's next move, which a hedge removes. The difference between a good and a bad counterparty shows up at the fill, in the half spread. The loss is carried by ten wallets, and the observables that the literature uses in place of identity do not find them.

Three of four pre-registered hypotheses are rejected, and the rejections are informative rather than empty. Toxicity is concentrated but not identifiable from trade size, and from sweeps only through the move of the underlying; under the registered rule, a net edge survives in 71.7 per cent of the cells and in short-dated at-the-money options, where H4 expected it to vanish; and an event study around the listing cannot tell an effect of this size from noise. What remains is a measurement, a fact about identity, and a map that says where on this surface a maker is paid, and how thinly.

Data, code and pre-registration

Every input to this paper is public. The trade tape and reference data come from the venue's public API, the volatility surfaces from the events of the feed contract on Derive Chain. The code that builds the sample, computes the markouts, runs the inference and draws every figure is at <https://github.com/gelatotrader/Derive-Research>, together with a test suite that covers it. The four hypotheses, their rejection rules and the inference were committed in docs/paper1/PRAEREGISTRERUNG.md on 17 September 2026 at 15:47 UTC+2, in commit cf1f432, before any markout had been computed. Two dated addenda resolved definitional questions that arose while the sample was being built, the

second after the first markouts and before the first inference; a third records the correction described on the first page, and a fourth the new commit hashes of 5 October 2026 (the registration was commit 3fd9caa before); all four are in the history of the same file. A reader who wants to check the order of events can read that history rather than take this paragraph on trust. Two registered robustness checks, a Deribit mark and a descriptive appendix on the nine other underlyings, are not reported in this version; every result the registration does not name is labelled exploratory.

Competing interest

I am building a market-making bot for the venue studied here and have been in contact with its team about obtaining order-level data for a follow-up paper. No data, funding or review was received from the venue for this paper, and every number in it comes from public sources. Readers should weigh the practitioner conclusions in Section 6 in that light.

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